# Arithmetic Restrictions on Mersenne Primes

A Self-Contained Note on OEIS A000668

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**Abstract**

We prove that the exponent of any Mersenne prime must be prime, and that every Mersenne prime other than 3 is congruent to 7 modulo 24. These are classical necessary conditions recorded in OEIS A000668, not a primality criterion or a solution to the open question of infinitely many Mersenne primes.

## Theorem and proof

**Theorem 1.** *If $M_n = 2^n - 1$ is prime for an integer $n \geq 2$, then $n$ is prime. Further, $M_n \equiv 7 \pmod{24}$ for every such prime with $n > 2$.*

*Proof.* If $n = ab$ with $a, b > 1$, then

$$2^n - 1 = (2^a)^b - 1 = (2^a - 1)\left(1 + 2^a + 2^{2a} + \cdots + 2^{a(b-1)}\right).$$

Both factors are integers greater than 1, so $2^n - 1$ is composite. Thus primality of $M_n$ forces $n$ to be prime. If $n > 2$ is prime, it is odd. Since $2^3 \equiv 0 \pmod 8$, we have $2^n - 1 \equiv 7 \pmod 8$. Since $2 \equiv -1 \pmod 3$ and $n$ is odd, $2^n - 1 \equiv (-1)^n - 1 \equiv 1 \pmod 3$. The unique residue modulo 24 that is 7 modulo 8 and 1 modulo 3 is 7. Hence $M_n \equiv 7 \pmod{24}$. □

## Boundary cases and limits

The first Mersenne prime is $M_2 = 3$, which is not $7 \pmod{24}$; it is explicitly excluded from the second assertion. The converse is false: a prime exponent does not ensure a Mersenne prime, e.g. $n = 11$ is prime but $2^{11} - 1 = 2047 = 23 \cdot 89$. The infinitude of Mersenne primes is not proved here.

## Scope and verification

Both assertions are classical and appear in the OEIS record; no novelty is claimed. The proof was written and checked by an AI without independent human peer review.

## References

1. OEIS Foundation, *A000668: Mersenne primes*, `https://oeis.org/A000668` (accessed 13 November 2026).
2. OEIS Foundation, *A000043: Mersenne prime exponents*, `https://oeis.org/A000043`.